

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE**Paper 2 (Generated practice paper - Set 05)**

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours**Materials**

For this paper you may use:

- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
 - use good English
 - organise information clearly
 - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
TOTAL	

Do not write outside the box

Answer all questions in the spaces provided.

0	1
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Table 1 compares organic and inorganic fertilisers.

0	1
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1

Complete Table 1 by giving two advantages and two disadvantages of each type of fertiliser.

[6 marks]

Table 1

Fertiliser	Advantages	Disadvantages
Organic (manure, sewage sludge, compost)		
Inorganic (eg ammonium nitrate, NPK)		

0	1
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2

Explain how hydroponic cultivation maximises productivity and give **two** disadvantages.

[4 marks]

10

0 2 . 1 Describe how each of the following threatens Antarctica.

[4 marks]

Global climate change

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Ozone depletion

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Tourism

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Future mineral exploitation

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0 2 . 2 Describe the main features of the Antarctic Treaty (1959) and the regulations that protect Antarctica.

[6 marks]

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10

03 . 1

Describe how cloning could be used to conserve the northern white rhinoceros, and explain why success has been limited.

[3 marks]

03 . 2

Explain how micro-propagation and cloning are used to conserve rare plants such as the Atlantic black poplar.

[3 marks]

03 . 3

Describe the purpose and methods of the Millennium Seed Bank.

[4 marks]

10

04 . 1

Describe the resources obtained from forests. Give **one** example for each category.

[5 marks]

Timber

.....

Fuel

.....

Food

.....

Fibres

.....

Medicines

.....

04 . 2

Explain why forests are an important carbon reservoir and how they can be used to counteract climate change.

[3 marks]

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0 4 . 3

Explain how forests regulate the hydrological cycle and local climate.

[2 marks]

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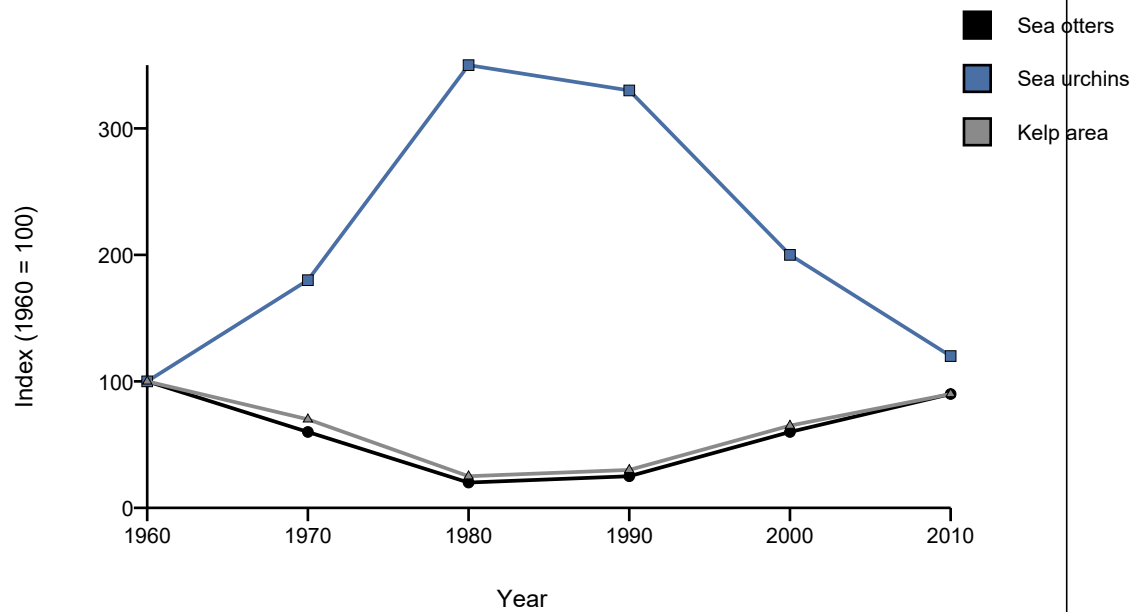
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10

0 5

Figure 1 shows the estimated populations of sea otters, sea urchins and the area of kelp forest along part of the coast of California between 1960 and 2010.

Figure 1



0 5 . 1

Describe the relationship between the three variables shown in Figure 1.

[3 marks]

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05 . 2

Explain the relationships shown in Figure 1.

[3 marks]

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05 . 3

Give **two** other examples of human over-exploitation of one species affecting the population of another.

[2 marks]

1

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2

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05 . 4

Explain why the loss of beavers or forest elephants can cause the decline of many other species.

[2 marks]

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10

06 . 1

State what is meant by **hybrid vigour**.

[1 mark]

06 . 2

Explain how cross-breeding Zebu and Ayrshire cattle produced a breed suited to hot climates.

[2 marks]

06 . 3

Give **two** potential applications of cloning to livestock farming.

[2 marks]

1

2

5

07 . 1

State the Brundtland Commission (1987) definition of sustainable development.

[1 mark]

07 . 2

Explain why activities such as slash-and-burn agriculture, fishing and sewage discharge can be sustainable with a small human population but not with a large one.

[3 marks]

07 . 3

Explain why urban populations are often less aware of unsustainable exploitation than rural communities.

[2 marks]

07 . 4

Describe **two** negative feedback mechanisms in natural systems and explain how human responses can override them.

[4 marks]

10

08 . 1

Explain why carnivorous species such as salmon dominate aquaculture in MEDCs while herbivores such as carp and tilapia dominate in LEDCs.

[3 marks]

08 . 2

Describe how all-female rainbow trout are produced and explain why this is done.

[3 marks]

08 . 3

Explain how each of the following abiotic factors is controlled in an intensive salmon farm.

[4 marks]

Dissolved oxygen

.....

Water flow

.....

Day length

.....

Temperature

.....

10

0 9

Table 1 shows some features of Earth, Mars and Venus.

Table 1

Feature	Venus	Earth	Mars
Mean distance from Sun / million km	108	150	228
Mass relative to Earth	0.82	1.0	0.11
Atmospheric pressure relative to Earth	92	1.0	0.006
Mean surface temperature / °C	460	15	-60
Magnetic field	Very weak	Strong	Very weak
Rotation period / hours	5 800	24	25

0 9**1**

Use Table 1 to explain why liquid water cannot exist on the surface of Mars.

[3 marks]

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0 9**2**

Explain how the Earth's mass and its distance from the Sun created suitable conditions for life.

[3 marks]

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0	9	.	3
---	---	---	---

Explain the importance of the Earth's magnetic field and its speed of rotation for living organisms.

[2 marks]

0	9	.	4
---	---	---	---

Explain how the tilt of the Earth's axis affects living organisms.

[2 marks]

10

1 0 . 1

Explain how an understanding of ecological succession and population dynamics can inform conservation decision-making.

[9 marks]

10 . 2

State what is meant by a **taxon**.

[1 mark]

.....

.....

10

Do not write outside the box

Write an essay on one of the following topics.

$$\boxed{1} \boxed{1} . \boxed{1}$$

Discuss the extent to which the resources and services provided by other species justify the conservation of biodiversity.

[25 marks]

OR

1 1 . 2

Discuss how an understanding of energy flow and the manipulation of food species can be used to increase food production sustainably.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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Do not write outside the box

END OF QUESTIONS